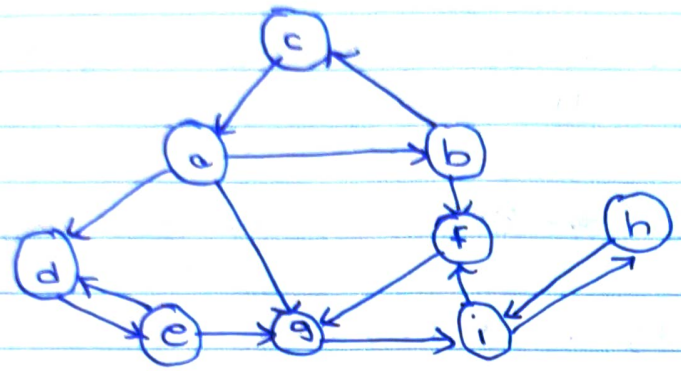
# PART - II

Problem 1:




Step 1 : Start at a . Outgoing edges : g, b, d

dt(a) = 1

Step 2 : (a, g) . Outgoing edge : i

dt(g) = 2

Step 3 : (g, i) . Outgoing edge : h

dt(i) = 3

Step 4 : (i, h) . No more edges move back to i

dt(h) = 4, Ft(h) = 5

Step 5 : Backtract to i & move to (i, f).

outgoing edges : g . dt(f) = 6

Step 6 : (f, g) . Since g has been visited

Ft(f) = 7

Step 7 : Backtrack to i . No more edges so

Ft(i) = 8

Step 8 : Backtrack to g . No more edges

Ft(g) = 9

Step 9: Backtrack to a . Move (a, d)
dt(d) = 10 .

Step 10: (d, e) . Outgoing edges : g, d
dt (e) = 11

Step 11: (e, g) This is an cross edge

Step 12: (e, d) . ft(e) = 12

Step 13: Backtrack to d . No more edges so
ft (d) = 13

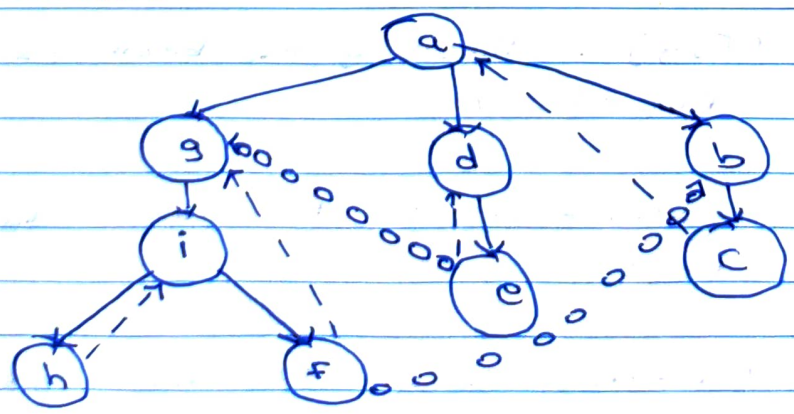
Step 14: Backtrack to a . Move to (a, b) .
dt (b) = 14 , Outgoing edges f, c

Step 15: (b, f) . It's Finished . (b, c)
dt (c) = 15 . outgoing edge : a

Step 16: (c, a) , ft (c) = 16

Step 17: Backtrack to b . No more edges . ft(b)=17
Backtrack to a . No more edges . ft(a)=18

Problem 2:




- - - : Backedge
——— : Tree edge
ooo : Crossedge

Tree edges:

(a, g), (g, i), (i, h), (i, f), (a, d), (d, e)
(a, b), (b, c)

Back edge:

(h, i), (f, g), (e, d), (c, a)

Cross edge:

(e, g), (b, f)

Problem 3:

| Vertex | Predecessor | Discovery Time | Finish Time |
|---|---|---|---|
| a | None | 1 | 18 |
| b | a | 14 | 17 |
| c | b | 15 | 16 |
| d | a | 10 | 13 |
| e | d | 11 | 12 |
| f | i | 6 | 7 |
| g | a | 2 | 9 |
| h | i | 4 | 5 |
| i | g | 3 | 8 |

(No AI was used)